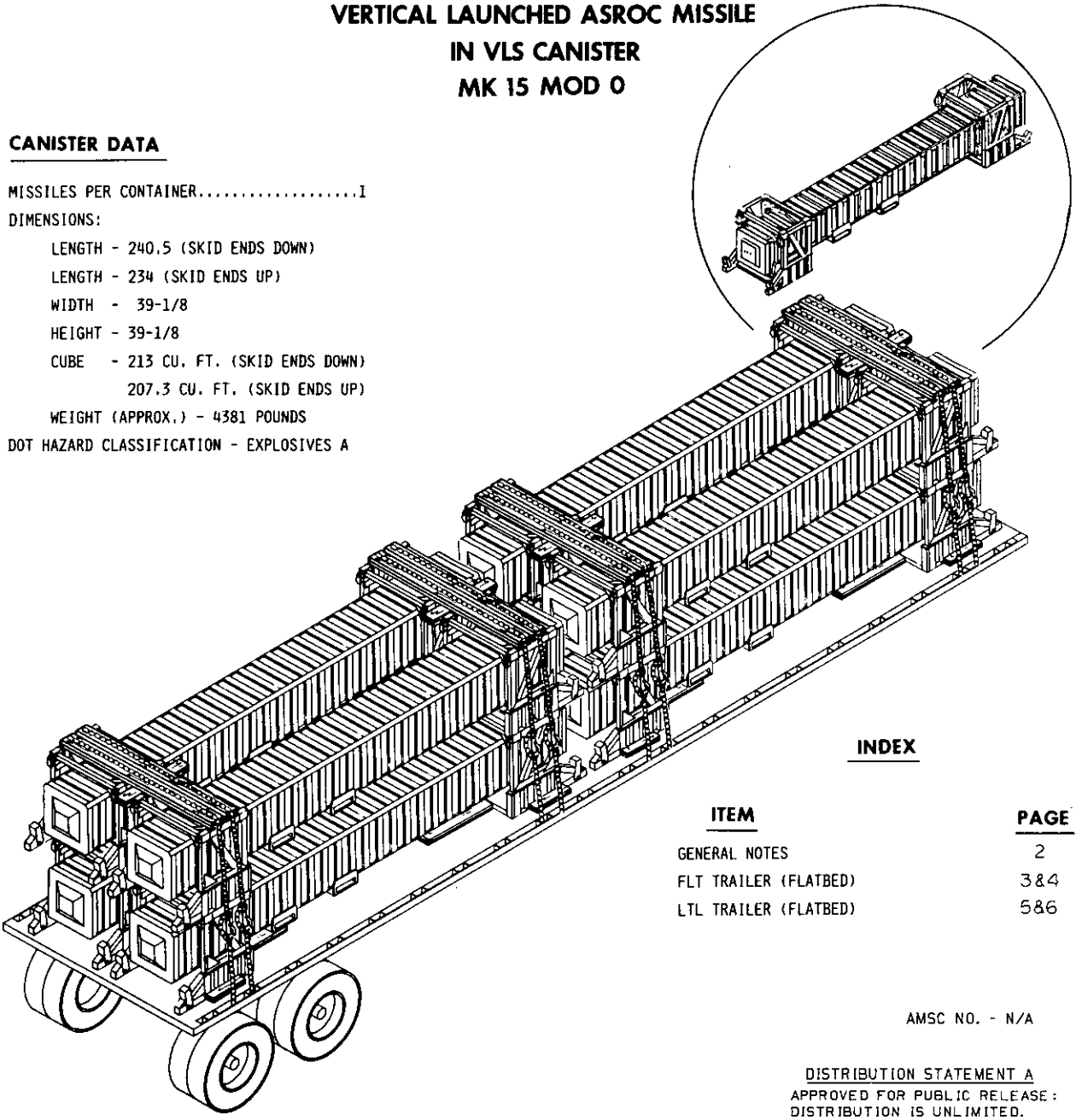
MILITARY STANDARD

MIL-STD-1320-248
(NAVY)
7 SEPTEMBER 1989

TRUCKLOADING
VERTICAL LAUNCHED ASROC MISSILE
IN VLS CANISTER
MK 15 MOD 0



CANISTER DATA

MISSILES PER CONTAINER.....1
DIMENSIONS:
- LENGTH - 240.5 (SKID ENDS DOWN)
- LENGTH - 234 (SKID ENDS UP)
- WIDTH - 39-1/8
- HEIGHT - 39-1/8
- CUBE - 213 CU. FT. (SKID ENDS DOWN)
  207.3 CU. FT. (SKID ENDS UP)
- WEIGHT (APPROX.) - 4381 POUNDS

DOT HAZARD CLASSIFICATION - EXPLOSIVES A

INDEX

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FLT TRAILER (FLATBED)	3&4
LTL TRAILER (FLATBED)	5&6

AMSC NO. - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE: DISTRIBUTION IS UNLIMITED.

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	

FSC 8140

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SIGNATURE	TDA, WPNSTA EARLE	DATE
		9-7-89
SIGNATURE	SEASYCOM BY DIRECTION	DATE
		1-7-89

ORIGINATOR 9/7/89 DATE

WPNSTA EARLE, NEW JERSEY

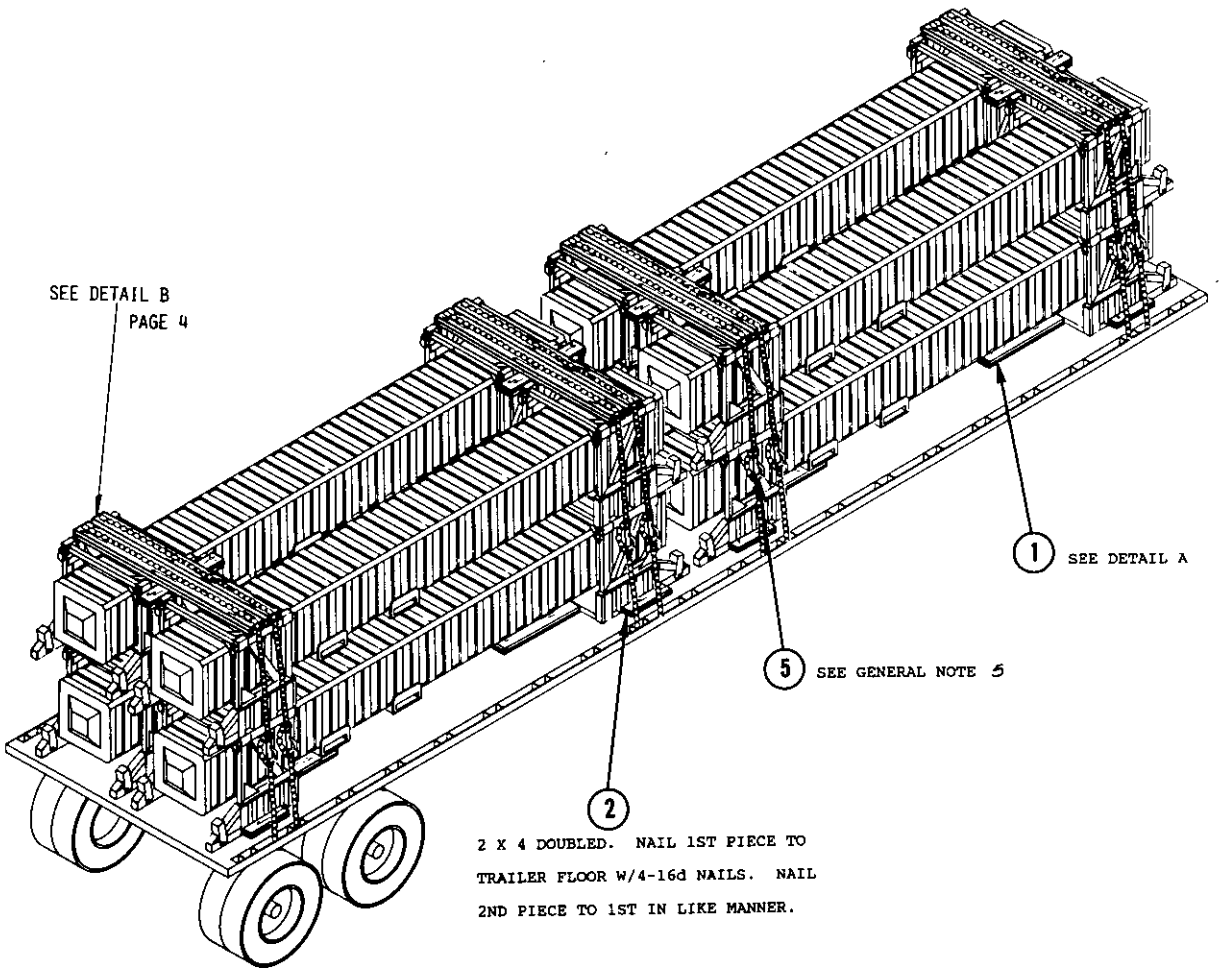
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GENERAL NOTES

1. THIS STANDARD PROVIDES DETAILED INSTRUCTION FOR TRUCKLOADING VLS CANISTERS LOADED WITH THE VL ASROC MISSILE. THE VLS CANISTERS SHALL BE EQUIPPED WITH ITS STACKING FRAME HARDWARE. THE SKID ENDS OF THE VLS CANISTER SHALL BE IN THE "UP" POSITION.
2. THE PROCEDURES AND PRACTICES CONTAINED HEREIN ARE INTENDED FOR 40 FT AND LONGER FLATBED TRAILERS. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS. TRAILER AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. HANDLE ALL CANISTERS ONE AT A TIME. DO NOT HANDLE TWO HIGH. WHEN THE CANISTERS ARE STACKED TWO HIGH, THE TOP CANISTER MAY BE BOLTED TO THE BOTTOM CANISTER BY BOLTING THE BOTTOM CANISTER STACKING FRAMES TO THE TOP CANISTER SKIDS USING THE BOLTS PROVIDED (4 BOLTS EACH END). BOLTING IS NOT MANDATORY.
4. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE TO ONE BASIS FOR TIEDOWNS.
5. STEEL STRAPPING USED AS TIEDOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
6. CHAINS, FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
7. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIEDOWNS. WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIEDOWN.
8. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION, REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
9. UNLESS OTHERWISE SPECIFIED DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER THE BLOCKING, BRACING AND TIEDOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIEDOWNS.
12. AFTER THE TARPAULINS ARE IN PLACE, ATTACH "EXPLOSIVES A" PLACARDS TO BOTH SIDES, FRONT AND BACK OF THE TRAILER AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
14. NYLON WEBBING MAY BE SUBSTITUTED FOR THE CHAIN AND LOAD BINDERS FOR SHIPMENTS OF EMPTY OR EXPENDED CANISTERS.

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FLT 40, 42, 44 AND 45 FT TRAILER (FLATBED)



TRUCKLOAD DATA

NUMBER OF CANISTERS 8

WEIGHT (APPROX.) ..... 35048 LBS.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	5	CHAIN AND BINDER		5/16 OR 3/8
4	4	SEPARATOR CLEAT	DETAIL B	2 X 4 X 36
16	3	PROTECTOR BOARD	DETAIL B	2 X 6 X 82
16	2	SLEEPER		2 X 4 X 18
32	1	BACKUP CLEAT		2 X 6 X 36

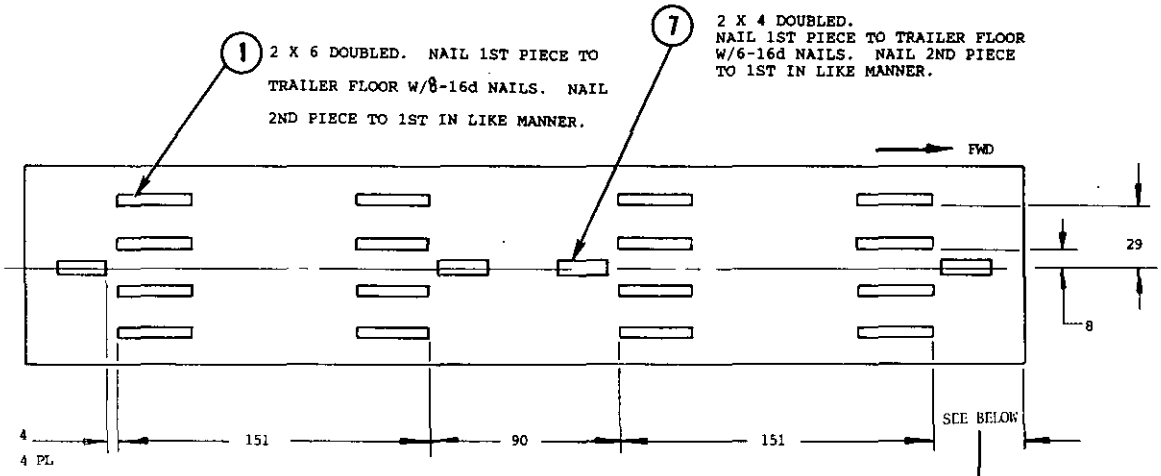
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FLT 40, 42, 44 AND 45 FT TRAILER (FLATBED)

1. A FULL TRUCKLOAD CONSISTS OF EIGHT VLS CANISTERS.
2. THE PREPOSITIONED BLOCKING SHOWN IN DETAIL A MUST BE INSTALLED ON THE TRAILER FLOOR BEFORE LOADING THE CANISTERS.

NOTE

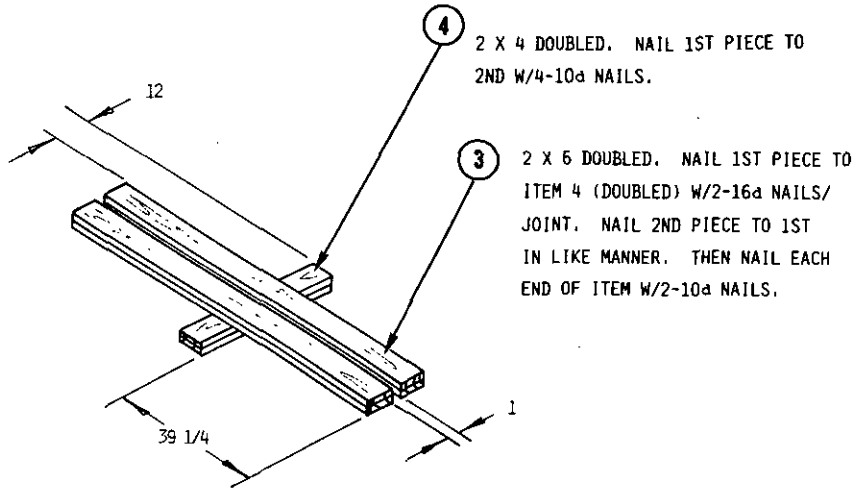
THE SKID ENDS OF THE CANISTERS MUST BE IN THE "UP" POSITION FOR CANISTERS TO FIT OVER THE PREPOSITIONED BLOCKING.



DETAIL A

PREPOSITIONED BLOCKING

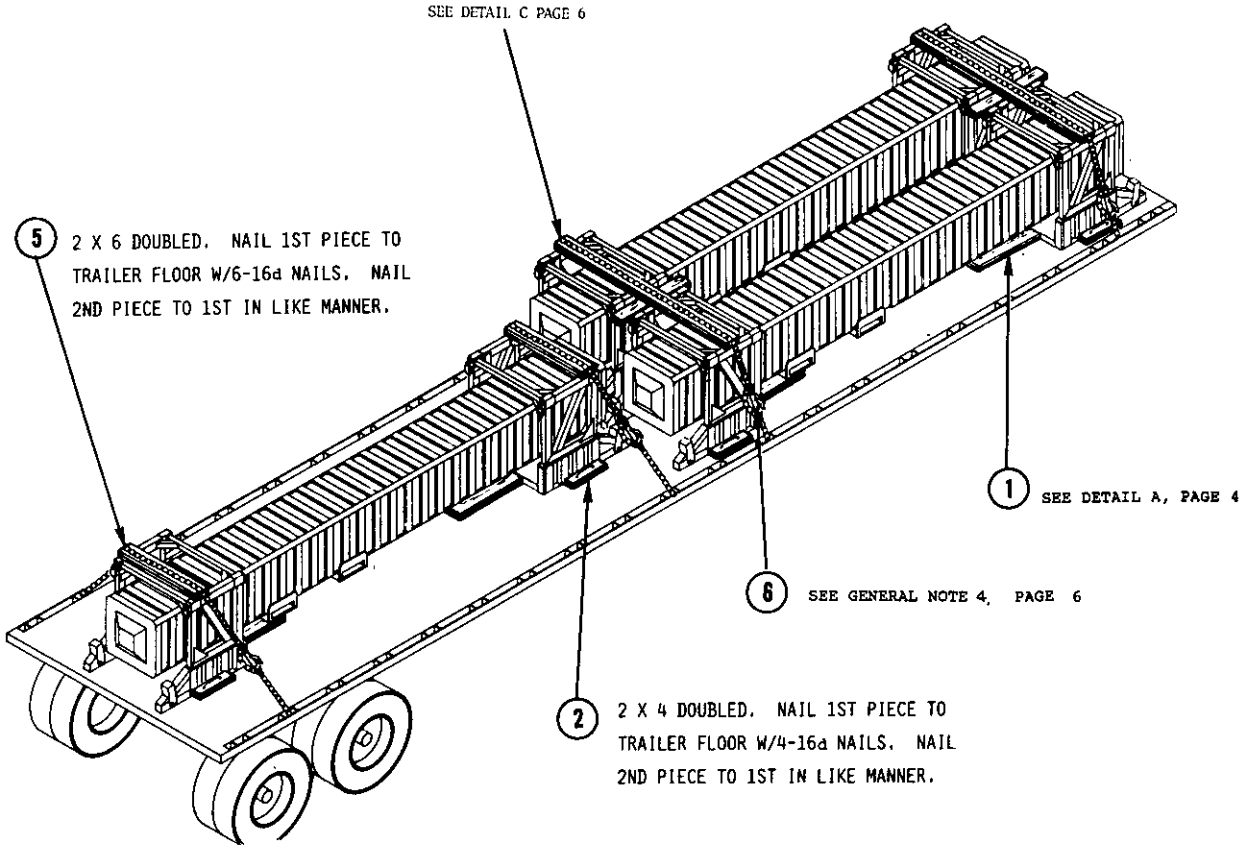
DIMENSION FOR 40 FT TRAILER	44
DIMENSION FOR 42 FT TRAILER	56
DIMENSION FOR 44 FT TRAILER	68
DIMENSION FOR 45 FT TRAILER	74



DETAIL B

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LTL 40, 42, 44 AND 45 FT TRAILER (FLATBED)



REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	7	SEPARATOR		2 X 4 X 24
4	6	CHAIN & BINDER		5/16 OR 3/8
4	5	PROTECTOR BOARD		2 x 6 x 40
4	4	SEPARATOR CLEAT	DETAIL C	2 x 4 x 36
4	3	PROTECTOR BOARD	DETAIL C	2 x 6 x 82
16	2	SLEEPER		2 x 4 x 18
24	1	BACKUP CLEAT		2 x 6 x 36

LIST OF MATERIALS

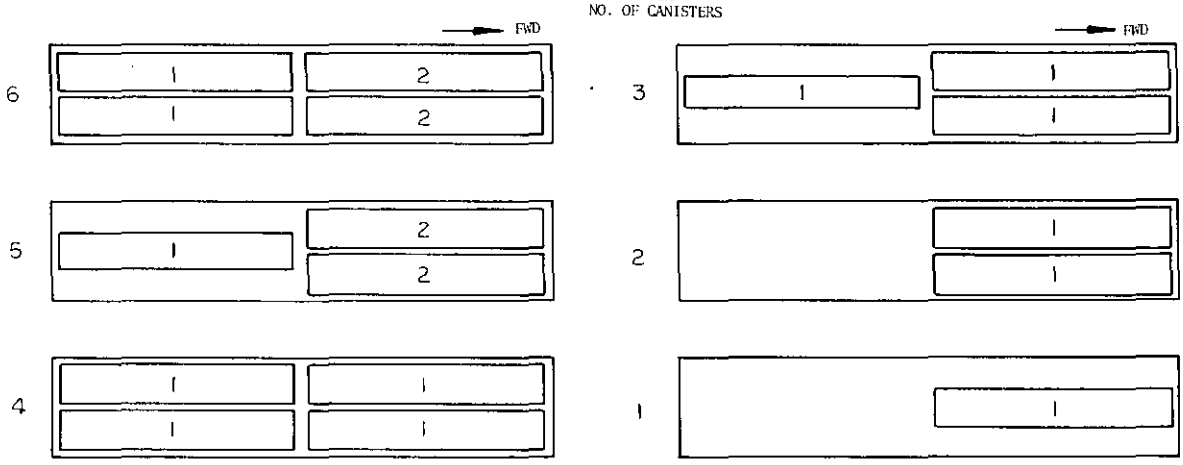
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LTL 40, 42, 44 AND 45 FT TRAILER (FLATBED)

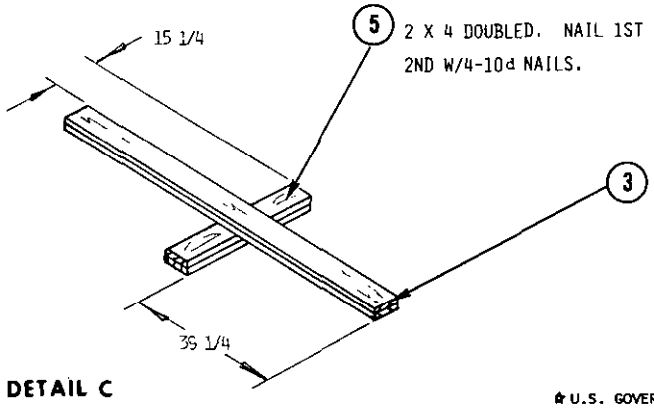
1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CANISTERS TO BE SHIPPED FROM TABLE I AND POSITION CANISTERS ON THE TRAILER ACCORDINGLY.
2. SEVEN CANISTERS CREATE A STACK UNEVEN IN HEIGHT AND SHALL NOT BE SHIPPED BECAUSE OF WEIGHT DISTRIBUTION AND TIEDOWN PROBLEMS.
3. BLOCK AND TIEDOWN LADING AS SHOWN ON PAGES 3 AND 5.
4. PROTECTOR BOARDS (DETAILS B AND C) AND CHAINS AND BINDERS (PIECE NO. 6) MUST BE POSITIONED ON STACKING FRAMES AS SHOWN ON DRAWING.
5. WHEN SHIPPING 3 OR 5 CANISTERS, CENTER THE PREPOSITIONED BLOCKING FOR THE SINGLE CANISTER ON THE CENTER LINE OF THE TRAILER.
6. WHEN SHIPPING 1 OR 2 CANISTERS, BLOCKING MAY BE ACCOMPLISHED BY USING 2 X 6 X 36 DOUBLED BACKUP CLEATS AT THE ENDS OF THE CANISTERS. USE TWO DOUBLED BACKUP CLEATS AT EACH END POSITIONED AGAINST THE LOWER CROSSMEMBER. FORWARD END OF THE CANISTERS SHALL BE 40 INCHES MINIMUM FROM FRONT OF TRAILER.
7. TRAILERS SHORTER THAN 40 FT MAY BE USED HOWEVER ONLY 4, 2 OR 1 CANISTERS CAN BE LOADED. USE THE PRINCIPLES OF THIS STANDARD TO BLOCK AND TIEDOWN. SHIPMENT MAY BE MADE ON "DOUBLES" WHEN STATE LAW PERMITS.
8. THE LTL PLAN SHOWN ON PAGE 5 DOES NOT LIMIT SHIPMENTS TO THIS QUANTITY. ITS PRESENTATION IS INTENDED TO EXPLAIN THE CORRECT METHODS OF LOADING SHIPMENTS.

TABLE I

NO. OF CANISTERS	*LOAD PATTERN NO. OF CANISTERS
6	3
5	2
4	1



* LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER. NUMBER INDICATES THE NUMBER OF CANISTERS IN A LAYER.



DETAIL C

* U.S. GOVERNMENT PRINTING OFFICE: 1990 704-034/30152

REVIEW ACTIVITY:
NAVY-OS

PREPARING ACTIVITY:
NAVY-SH
(PROJECT NO. 8140-N835)

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